SC-LAB at ALTA 2026: Variety-Robust Sentiment and Sarcasm Classification through Per-Cell Ensemble Selection

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Abstract

We present the SC-LAB submission to the ALTA 2026 shared task on joint sentiment and sarcasm classification for Australian and British English. The official metric scores each sub-task by its weaker variety, so variety robustness rather than average accuracy determines the final result. Our system targets this structure directly: we train a pool of 17 transformer models, adapt them to the sarcasm domain by intermediate pretraining on 250k Reddit comments, and combine them using ensemble selection performed independently for each (task, variety, source) cell, with decision thresholds fitted on out-of-fold predictions. The system scores **0.8526** on the blind test set, and **0.9304** on the development target, where it ranked first. We then characterise the relationship between the two evaluation phases, showing that development scores are optimistic by approximately 0.08 because the protocol permits fitting against a target with public labels, whereas nested cross-validation predicts blind-test performance to within 0.01. Per-cell ablations identify intermediate pretraining as the dominant contributor, and we report four negative results on ensembling that we expect to be useful to future participants.

1 Introduction

Sentiment classification is a standard sequence-classification problem, but performance degrades unevenly across national varieties of English. The ALTA 2026 shared task evaluates joint sentiment and sarcasm classification over the en-AU and en-UK subsets of BESSTIE, drawn from Google Places reviews and Reddit comments, and scores each sub-task by its *weaker* variety, so a system that is strong on average but uneven across varieties gains nothing from its stronger side. We therefore treat the task as four independent classification problems. Our contributions are:

- A per-(task, variety, source) ensemble selection procedure with out-of-fold threshold tuning, which treats each scored cell as its own optimisation target (§3).
- A domain-adaptive pretraining study showing that scaling an intermediate Reddit sarcasm corpus from 60k to 250k examples improves pooled out-of-fold sarcasm en-UK by 0.037, the single largest gain we measured (§3.2).
- An analysis of the two evaluation phases demonstrating that development-phase scores are inflated by approximately 0.08, together with a nested cross-validation protocol that predicted our blind-test score to within 0.01 (§5).
- Four negative results on ensembling and augmentation, each measured against a matched baseline (§5).

2 Task and Data

The official metric averages each sub-task's weaker variety:

$$\text{score} = \tfrac{1}{2}\big(\min(F_1^{\text{s-AU}}, F_1^{\text{s-UK}}) + \min(F_1^{\text{r-AU}}, F_1^{\text{r-UK}})\big)$$

where each F_1 is macro-F_1 over a (task, variety) cell, s denotes sentiment and r denotes sarcasm. Capacity spent on a variety that is not the minimum is wasted, and a single ensemble tuned on pooled accuracy optimises an average the metric never evaluates. Both observations motivate the per-cell design in §3.

The data comprises 5,531 training and 757 validation rows, with a blind test release of 1,573 rows. Each row carries `source` (Google or Reddit), `variety`, `text`, and binary `sentiment` and `sarcasm` labels. Both `source` and `variety` are observed at inference time, so conditioning thresholds on them is legitimate.

source	variety	n	sent. pos.	sarc. pos.
google	en-AU	946	0.735	0.070
google	en-UK	1,817	0.748	**0.001**
reddit	en-AU	1,763	0.319	0.419
reddit	en-UK	1,005	0.114	0.217

Table 1: Training class distribution. Sarcasm is near-absent from Google reviews and effectively nonexistent in google/en-UK (≈2 rows).

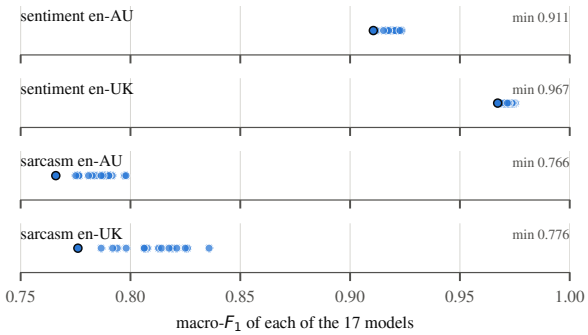


Figure 1: Each dot is one of the 17 models on one scored cell; the ringed dot is that cell's weakest model. Sentiment is tightly clustered and near-solved, while sarcasm sits 0.13–0.15 lower and is far more dispersed. Because the metric keeps only the weaker variety per task, the two sarcasm panels decide the score almost entirely.

Table 1 shows the central difficulty. Sarcasm is almost absent from Google reviews, yet sarcasm-en-UK supplies half the second term of the score while resting on 218 Reddit positives, so one component of the metric is determined by a few hundred examples. Figure 1 shows the effect across our pool: sentiment is effectively solved while both sarcasm cells lie far below it. We additionally observe $P(\text{negative} \mid \text{sarcastic}) = 96.7\%$ in training, indicating that annotators recorded intended meaning and that the two tasks are mutually informative.

3 System Description

3.1 Base Models

Each model is a shared encoder, mean-pooled over the attention mask, with two independent 2-class heads trained under a summed loss carrying a tunable multiplier on the sarcasm term. Every input is prefixed with its `source` and `variety`, both observed at training and test time. Inputs use head+tail truncation at 192 tokens (75%/25%), class-balanced loss weights, label smoothing of 0.05, multi-sample dropout, and AdamW with layer-wise learning-rate decay. Encoders are fully fine-tuned; 7–8B decoders use LoRA adapters (Hu et al., 2022) with r=16 over 4-bit NF4 weights (Dettmers et al., 2023) and last-token pooling. All models use 5-fold cross-validation stratified jointly on source × variety × sentiment × sarcasm, so that cells with single-digit counts cannot fall entirely in one fold.

intermediate corpus	r-AU	r-UK	r-UK (OOF)
60k Reddit, 1 epoch	0.7837	0.8002	0.8097
250k Reddit, 1 epoch	**0.7895**	**0.8191**	**0.8463**

Table 2: Scaling the intermediate pretraining corpus. The first two columns are fold means; the third is pooled out-of-fold sarcasm macro-F_1.

The final pool of 17 models comprises 14 DeBERTa-v3-large (He et al., 2021) variants (seeds 42/1337/2024; sarcasm-weighted; focal loss; variety-augmented; and the intermediate-pretraining variants described below), 2 RoBERTa-large (Liu et al., 2019), and Llama-3.1-8B-Instruct (Llama Team, AI @ Meta, 2024).

3.2 Domain-Adaptive Pretraining

This produced our largest single gain, and is an instance of domain-adaptive pretraining (Gururangan et al., 2020). Every sarcastic row in BESSTIE is a Reddit comment, but our initial intermediate-task pretraining used news headlines and tweets — a domain mismatch on precisely the task that binds the metric. We built an 847k-row Reddit sarcasm corpus from SARC (Khodak et al., 2018) and pretrained the encoder on a 250k balanced subsample before task fine-tuning. Table 2 compares this against the smaller 60k corpus on matched fold splits.

3.3 Per-Cell Ensemble Selection

Selection operates on cached out-of-fold probability arrays rather than raw text, making the search space cheap to explore. Probabilities are rank-normalised before averaging, since the pool mixes bidirectional encoders with causal decoders whose probability scales differ (raw-probability averaging 0.8356 versus rank averaging 0.8670). Each cell receives its own ensemble, and within a cell each `source` its own weights and threshold.

We compared four combination rules, all variants of ensemble selection from a library of models (Caruana et al., 2004): unweighted rank averaging; greedy subset selection; a weighted blend over an expanded feature space, in which each model contributes its rank, its raw probability, its cross-

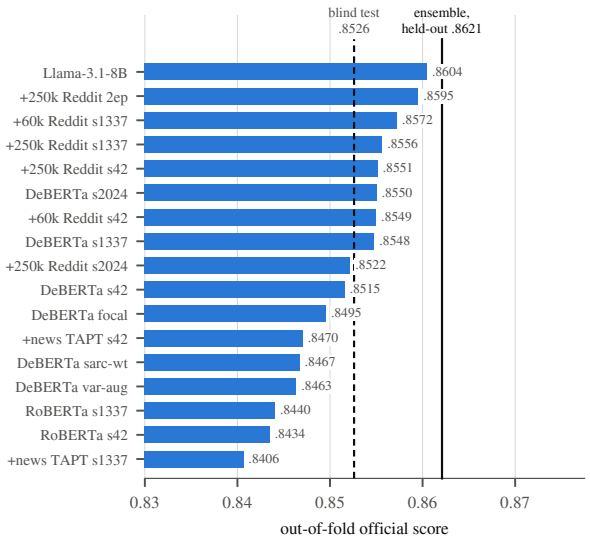


Figure 2: All 17 pool members, out-of-fold over the 6,288-row pool at a fixed 0.5 threshold, against the ensemble built from them. The best single model (.8604) reaches within .0017 of the ensemble's held-out estimate and exceeds its realised blind-test score: combination contributes far less here than model quality does.

system	s-AU	s-UK	r-AU	r-UK	score
expanded blend	.9566	.9870	.8934	.8978	.9250
+ re-opt. sent.	**.9675**	**.9948**	.8934	.8978	**.9304**

Table 3: Development-phase results, fitted against the development target and scored on that same target.

task rank and its inverted rank, with a per-source choice of mean/median/max/min aggregation; and bagged greedy selection over bootstrap resamples (Caruana et al., 2006).

For the test phase all 17 models were retrained on the union of the training and validation splits (6,288 rows, +14% data, raising en-UK sarcasm positives from 218 to 248), and every weight, aggregation and threshold was fitted on out-of-fold predictions only.

4 Results

4.1 Development Phase

Under the task's development protocol, ensemble weights and thresholds may be fitted against the 757-row development target. Table 3 reports our submission and its improvement.

A later 2,000-restart search raised sarcasm-en-UK to 0.9123 but left the binding sarcasm-en-AU cell at 0.8934, so the final score did not move at all — a direct demonstration that only the argmin matters.

combination rule	held-out	in-samp.	gap
rank averaging	.8568 ±.0037	.8629	.006
greedy subset	**.8621** ±.0076	.8753	.013
bagged greedy	.8615 ±.0050	—	—
expanded blend	.8578 ±.0067	.8788	.021

Table 4: Test-phase combination rules, all fitted on out-of-fold predictions with no access to test labels.

4.2 Test Phase

All weights and thresholds were fitted on out-of-fold predictions. The held-out column of Table 4 is nested cross-validation: the blend is selected on 4/5 of the out-of-fold rows and scored on the withheld 1/5.

We submitted greedy subset selection. It scored **0.8526** on the blind test set, placing 6th. Our pre-submission estimate was 0.8621, so the prediction was accurate to 0.0095 — consistent with ordinary selection optimism.

5 Discussion

Development scores are inflated by ≈0.08. Development submissions are scored against a file whose gold labels are distributed with the public data, so fitting ensemble weights and thresholds against it measures training fit rather than generalisation. Our 0.9304 was obtained under that protocol; the identical pipeline refitted on out-of-fold predictions scores 0.8526 on blind data. A simple diagnostic detects the effect: our best development-phase sarcasm cell (0.8329) exceeds the oracle-on-target ceiling of every individual pool member (0.8162), which is unattainable without target information. We recommend that future editions either withhold development labels or report a held-out estimate alongside the leaderboard figure.

Nested cross-validation is well calibrated. It predicted 0.8621 against a realised 0.8526. In-sample scores are not predictive: the expanded-feature blend attains the best in-sample score (0.8788) and the second-worst held-out score, a gap of 0.021 (Table 4).

Effective components. Scaling intermediate pre-training contributes most (+0.037 pooled out-of-fold on sarcasm-en-UK), followed by per-cell ensembling. Two models with unremarkable aggregate scores carry the largest weight in specific cells, confirming that members must be assessed per cell. Restricting sarcasm-specialised members to the sarcasm cells is necessary: including them in all four

regresses sentiment-en-AU from 0.9321 to 0.9266 and cancels the sarcasm gain.

Negative results. The following were measured against matched baselines and rejected. *Variety-invariance augmentation*: randomly neutralising the variety token to force a single shared detector, motivated by en-UK having 218 sarcastic training rows against en-AU's 739, gave out-of-fold sarc-AU $0.7896 \rightarrow 0.7767$ and sarc-UK $0.8160 \rightarrow 0.8140$ — worse on both, refuting the hypothesis. *Bagged selection*, the one methodological gap in our submitted rule, changed nothing (0.8615 versus 0.8621), indicating the variance was not in the selection step. *Prior-matched quantile thresholding* (0.8537) and *logistic stacking* with a cross-task sentiment feature (0.8591) both fell below plain argmax thresholding (0.8611). *A second seed* of our best pretrained model collapsed sarc-UK to 0.7553: with roughly 30 validation positives deciding that cell, adding correlated members is actively dangerous.

Ensembling saturated. Across five combination rules the held-out spread was only 0.0053, and pairwise disagreement between their test-set predictions was 1% for sentiment and 3–4% for sarcasm. As Figure 2 shows, the best single model (0.8604 out-of-fold) came within 0.0017 of the full 17-model blend's held-out estimate and in fact exceeded its realised blind-test score. The remaining headroom is in the base models, not the combiner.

6 Practical Considerations

Training is sensitive to library versions in a way that produces no error signal. Under Python 3.13 with PyTorch 2.12 and Transformers 5.11, DeBERTa-v3-large diverges: training loss *rises* from 1.28 to 1.42, chance level for two binary tasks, whereas under Python 3.8 with PyTorch 2.1.2 and Transformers 4.46.3 it falls from 0.99 to 0.50 on identical data, seed and hardware. No exception or warning is raised. The older environment, however, cannot load Qwen3 (which requires Transformers ≥ 4.51) and rejects Phi-4-mini's `rope_scaling` configuration, so the pool composition is itself constrained by the environment. We therefore pin the training environment and assert that first-epoch loss falls below a known-good threshold before committing compute to a full run.

7 Conclusion

We described a per-cell ensemble system for variety-robust sentiment and sarcasm classification, scoring 0.8526 on the ALTA 2026 blind test set and 0.9304 on the development target. Domain-adaptive pretraining on Reddit sarcasm is the largest single contributor. We showed that development and blind-test scores are not comparable under this task's protocol, and that nested cross-validation closes the gap to 0.01. Since five combination rules span only 0.005 while the best single model nearly matches the full ensemble, we expect further gains to come from stronger base models and from the $\approx$250 en-UK sarcasm positives that determine the binding cell, rather than from the combiner.

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